

Relatório RBF

Eduardo

April 19, 2025

Introduction

This report presents the implementation and evaluation of a Radial Basis Function (RBF) classifier with L2 regularization. The model leverages K-means clustering for center selection and employs regularized least squares for weight optimization. Experiments were conducted on synthetic and real-world datasets to validate performance.

Methodology

RBF Network Structure

The classifier computes activations using the RBF kernel:

$$\phi(\mathbf{x}) = \exp(-\gamma \|\mathbf{x} - \mathbf{c}_i\|^2) \quad (1)$$

where $\mathbf{c}_i$ are cluster centers and γ is determined by:

$$\gamma = \frac{1}{2 \cdot (\text{median}(\text{min_dists}))^2} \quad (2)$$

with fallback to data-to-center distances if centers collapse.

Weight Optimization

Weights $\mathbf{W}$ are solved using regularized least squares:

$$\mathbf{W} = (\mathbf{H}^T \mathbf{H} + \lambda \mathbf{I})^{-1} \mathbf{H}^T \mathbf{Y} \quad (3)$$

where $\mathbf{H}$ is the RBF activation matrix and λ is the regularization parameter.

Implementation Details

- Centers selected via K-means clustering
- Automatic γ computation using median heuristic
- L2 regularization for numerical stability
- One-vs-rest strategy for multi-class support

Experiments

Synthetic Data

- 1,000 samples, 20% test split
- Default parameters: 100 centers, $\lambda = 10^{-3}$

Breast Cancer Dataset

- Preprocessing: Drop correlated features
- 10-fold cross-validation
- Evaluation metric: Classification accuracy

Results

Synthetic Data Evaluation

$$\text{Accuracy} = 0.88$$

Cross-Validation Performance

Mean Accuracy	0.89
Std Deviation	0.03

Table 1: 10-fold cross-validation results on breast cancer dataset

Detailed fold scores: [0.91, 0.89, 0.88, 0.95, 0.86, 0.88, 0.84, 0.88, 0.88, 0.95]

Conclusion

The RBF classifier demonstrated strong performance on both synthetic (88% accuracy) and real-world data (89% mean cross-validated accuracy). The median-based γ computation and L2 regularization proved effective for stable learning.